GENOME ASSEMBLY
Barcode Project

Abane Louis Ashu

Setting Up The Working Environment

abane_ashu@AbaneAshu: /mnt/c/Windows/system32

```
Windows Subsystem for Linux is now available in the Microsoft Store!
You can upgrade by running 'wsl.exe --update' or by visiting https://aka.ms/wslstorepage
Installing WSL from the Microsoft Store will give you the latest WSL updates, faster.
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Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 4.4.0-19041-Microsoft x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Aug 11 18:21:51 CEST 2025

  System load:           0.52
  Usage of /home:        unknown
  Memory usage:          89%
  Swap usage:            6%
  Processes:             8
  Users logged in:       0
  IPv4 address for wifi0: 192.168.0.240
  IPv6 address for wifi0: 2a02:1811:c1c:2e00:fb01:c84e:4245:e2cb
  IPv6 address for wifi0: 2a02:1811:c1c:2e00::18
  IPv6 address for wifi0: 2a02:1811:c1c:2e00:cdb9:9b02:5564:7a79

This message is shown once a day. To disable it please create the
/home/abane_ashu/.hushlogin file.
(base) abane_ashu@AbaneAshu:/mnt/c/Windows/system32$
```

```
(base) abane_ashu@AbaneAshu:~$ pwd
/home/abane_ashu
(base) abane_ashu@AbaneAshu:~$ mkdir -p ~/miniconda3
(base) abane_ashu@AbaneAshu:~$ wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
--2025-08-11 19:26:25--  https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
Resolving repo.anaconda.com (repo.anaconda.com)... 2606:4700::6810:20f1, 2606:4700::6810:bf9e, 104.16.191.158, ...
Connecting to repo.anaconda.com (repo.anaconda.com)|2606:4700::6810:20f1|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 160039710 (153M) [application/octet-stream]
Saving to: ‘Miniconda3-latest-Linux-x86_64.sh.1’

Miniconda3-latest-Linux-x86_64.sh.1                 100%[===================================================================================================>] 152.62M  1.90MB/s    in 53s

2025-08-11 19:27:19 (2.87 MB/s) - ‘Miniconda3-latest-Linux-x86_64.sh.1’ saved [160039710/160039710]

(base) abane_ashu@AbaneAshu:~$ bash Miniconda3-latest-Linux-x86_64.sh

Welcome to Miniconda3 py313_25.5.1-0

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```

Raw Reads QA

```
## Raw reads check (seqkit)
**Command:** seqkit stats -a raw/barcode07.fq
```
file   format type num_seqs   sum_len min_len avg_len max_len   Q1   Q2   Q3 sum_gap   N50 N50_num Q20(%) Q30(%) AvgQual GC(%) sum_n
raw/barcode07.fq FASTQ   DNA   37,054 392,624,046   462   10,596 165,410 3,525 4,294 10,991   0 24,577   4,324   83.98   57.02   16.96 41.29   0
```

num_seqs = 37,054
sum_len  = 392,624,046 bp  (~392.6 Mb)
avg_len  = 10,596 bp
N50      = 24,577 bp
Q20(%)   = 83.98
Q30(%)   = 57.02
GC(%)    = 41.29
```

```
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ echo "[Flye]"
[Flye]
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ seqkit stats -a ~/flot_project/assemblies/flye/assembly.fasta
file                                                      format  type  num_seqs    sum_len  min_len      avg_len    max_len     Q1           Q2         Q3  sum_gap       N50  N50_num  Q20(%)  Q30(%)  AvgQu
al  GC(%)  sum_n
/home/abane_ashu/flot_project/assemblies/flye/assembly.fasta  FASTA   DNA         2  3,780,699    7,344  1,890,349.5  3,773,355  7,344  1,890,349.5  3,773,355        0  3,773,355        1       0       0
 0  39.08      0
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ echo "[Raven]"
[Raven]
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ seqkit stats -a ~/flot_project/assemblies/raven/raven.fasta
file                                                     format  type  num_seqs    sum_len  min_len    avg_len    max_len      Q1      Q2       Q3  sum_gap       N50  N50_num  Q20(%)  Q30(%)  AvgQual  GC(%)
 sum_n
/home/abane_ashu/flot_project/assemblies/raven/raven.fasta  FASTA   DNA        28  6,402,553   32,563  228,662.6  3,773,352  60,518  81,406  162,482        0  3,773,352        1       0       0        0  46.73
     0
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ echo "[Shasta]"
[Shasta]
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ echo "[Shasta]"
[Shasta]
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$ seqkit stats -a ~/flot_project/assemblies/shasta/Assembly.fasta
file                                                        format  type  num_seqs    sum_len  min_len    avg_len  max_len      Q1       Q2       Q3  sum_gap      N50  N50_num  Q20(%)  Q30(%)  AvgQual  GC(%)
  sum_n
/home/abane_ashu/flot_project/assemblies/shasta/Assembly.fasta  FASTA   DNA        13  3,155,007   10,492  242,692.8  552,786  85,120  234,105  437,312        0  437,312        4       0       0        0  39.09
      0
(assem) abane_ashu@AbaneAshu:~/flot_project/eval/16s$
```

Running The Assemblers

Flye assembly
Command: flye --nano-hq raw/barcode07.fq --out-dir assemblies/flye --threads $(nproc)

```
total 9.3M
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:50 00-assembly
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:54 10-consensus
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:55 20-repeat
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 20:55 30-contigger
drwxr-xr-x 1 abane_ashu abane_ashu  512 Aug 12 21:01 40-polishing
-rw-r--r-- 1 abane_ashu abane_ashu 3.7M Aug 12 21:01 assembly.fasta
-rw-r--r-- 1 abane_ashu abane_ashu 5.5M Aug 12 21:01 assembly_graph.gfa
-rw-r--r-- 1 abane_ashu abane_ashu 1.5K Aug 12 21:01 assembly_graph.gv
-rw-r--r-- 1 abane_ashu abane_ashu  120 Aug 12 21:01 assembly_info.txt
-rw-r--r-- 1 abane_ashu abane_ashu  45K Aug 12 21:01 flye.log
-rw-r--r-- 1 abane_ashu abane_ashu   92 Aug 12 21:01 params.json
```

Contigs & stats

```
contigs: 2
total_len = 3,780,699 bp
N50       = 3,773,355 bp
largest   = 3,773,355 bp
mean_cov  = 85×
```

Raven assembly
Command:

```
raven \
  --threads $(nproc) \
  --graph assemblies/raven/raven.gfa \
  raw/barcode07.fq \
  > assemblies/raven/raven.fasta \
  2> logs/raven_withgraph_$(date +%F_%H%M).log
```

Output files:

```
total 15M
-rw-r--r-- 1 abane_ashu abane_ashu 6.2M Aug 12 21:51 raven.fasta
-rw-r--r-- 1 abane_ashu abane_ashu 7.8M Aug 12 21:51 raven.gfa
```

Contigs & stats:
Command:

```
grep -c '^>' assemblies/raven/raven.fasta
seqkit stats -a assemblies/raven/raven.fasta
```

Results:

```
contigs   = 28
sum_len   = 6,402,553 bp
min_len   = 32,563 bp
avg_len   = 228,663 bp
max_len   = 3,773,352 bp
N50       = 3,773,352 bp
GC(%)     = 46.73
```

Shasta assembly
Command:

```
shasta \
  --input raw/barcode07.fq \
  --assemblyDirectory assemblies/shasta \
  --threads $(nproc) \
  --config Nanopore-May2022 \
  2>&1 | tee logs/shasta_$(date +%F_%H%M).log
```

Contigs & stats:
Command:

```
grep -c '^>' assemblies/shasta/Assembly.fasta
seqkit stats -a assemblies/shasta/Assembly.fasta
```

Results:

```
contigs   = 13
sum_len   = 3,155,007 bp
min_len   = 10,492 bp
avg_len   = 242,693 bp
max_len   = 552,786 bp
N50       = 437,312 bp
GC(%)     = 39.09
```

QUAST/N50

```
cd ~
wget https://github.com/ablab/quast/releases/download/quast_5.2.0/quast-5.2.0.tar.gz
tar -xzf quast-5.2.0.tar.gz
cd quast-5.2.0
python3 quast.py \
  -o ~/flot_project/eval/quast \
  -t $(nproc) \
  --labels Flye,Raven,Shasta \
  ~/flot_project/assemblies/flye/assembly.fasta \
  ~/flot_project/assemblies/raven/raven.fasta \
  ~/flot_project/assemblies/shasta/Assembly.fasta \
  2>&1 | tee ~/flot_project/logs/quast_$(date +%F_%H%M).log
```

Key metrics from report.tsv:

```
Assembler   #contigs(>=500bp)   N50        L50   Total_length   GC%
Flye        2                   3,773,355  1     3,780,699      39.08
Raven       28                  3,773,352  1     6,402,553      46.73
Shasta      13                    437,312  4     3,155,007      39.09
```

Raw grep outputs from report.tsv:
Command:

```
grep -P 'Assembly|# contigs (>= 500 bp)' eval/quast/report.tsv
```

Result:

```
Assembly        Flye    Raven   Shasta
```

Command:

```
grep \"contigs\" eval/quast/report.tsv
```

```
=== QUAST RESULTS ===
(assem) abane_ashu@AbaneAshu:~$ cat ~/flot_project/eval/quast/report.txt | grep -E "N50|Total length|# contigs"
All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).
# contigs (>= 0 bp)         2          28         13
# contigs (>= 1000 bp)      2          28         13
# contigs (>= 5000 bp)      2          28         13
# contigs (>= 10000 bp)     1          28         13
# contigs (>= 25000 bp)     1          28         12
# contigs (>= 50000 bp)     1          23         11
Total length (>= 0 bp)      3780699    6402553    3155007
Total length (>= 1000 bp)   3780699    6402553    3155007
Total length (>= 5000 bp)   3780699    6402553    3155007
Total length (>= 10000 bp)  3773355    6402553    3155007
Total length (>= 25000 bp)  3773355    6402553    3144515
Total length (>= 50000 bp)  3773355    6209102    3110678
# contigs                  2          28         13
Total length                3780699    6402553    3155007
N50                         3773355    3773352    437312
(assem) abane_ashu@AbaneAshu:~$ conda deactivate
```

KAT Analysis

Flye assembly — KAT results

Command:

```
kat comp \
  -o ~/flot_project/eval/kat_flye \
  ~/flot_project/raw/barcode07.fq \
  ~/flot_project/assemblies/flye/assembly.fasta
```

Key statistics:

```
Total k-mers in raw reads:        391,660,642
Total k-mers in assembly:         3,780,647
Distinct k-mers in raw reads:     77,765,317
Distinct k-mers in assembly:      3,725,038
Only in raw reads (distinct):     74,040,427
Only in assembly (distinct):      148
Shared distinct k-mers:           3,724,890
Estimated genome size:            3.77 Mb
Estimated heterozygosity:         0.04%
Assembly completeness:            99.85%
```

Raven assembly — KAT results

Command:

```
kat comp \
  -o ~/flot_project/eval/kat_raven \
  ~/flot_project/raw/barcode07.fq \
  ~/flot_project/assemblies/raven/raven.fasta
```

Key statistics:

```
Total k-mers in raw reads:        391,660,642
Total k-mers in assembly:         6,401,825
Distinct k-mers in raw reads:     77,765,317
Distinct k-mers in assembly:      6,316,936
Only in raw reads (distinct):     71,456,286
Only in assembly (distinct):      7,905
Shared distinct k-mers:           6,309,031
Estimated genome size:            3.77 Mb
Estimated heterozygosity:         0.04%
Assembly completeness:            99.90%
```

- Genome size matches expected value, no large-scale duplications detected.

Shasta assembly — KAT results

Command:

```
kat comp \
  -o ~/flot_project/eval/kat_shasta \
  ~/flot_project/raw/barcode07.fq \
  ~/flot_project/assemblies/shasta/Assembly.fasta
```

Key statistics:

```
Total k-mers in raw reads:        391,660,642
Total k-mers in assembly:         3,154,669
Distinct k-mers in raw reads:     77,765,317
Distinct k-mers in assembly:      3,112,375
Only in raw reads (distinct):     74,655,961
Only in assembly (distinct):      3,019
Shared distinct k-mers:           3,109,356
Estimated genome size:            3.77 Mb
Estimated heterozygosity:         0.04%
Assembly completeness:            88.41%
```

BUSCO Evaluation

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```
wget https://gitlab.com/ezlab/busco/-/archive/5.8.0/busco-5.8.0.tar.gz
tar -xzf busco-5.8.0.tar.gz
cd busco-5.8.0
conda install -y pandas numpy matplotlib biopython hmmer blast augustus prodigal sepp openmpi ucx
conda install -y -c bioconda bbmap
pip install requests
pip install .
```

Note: Each installation step was tested by running `busco --help` to confirm functionality before proceeding.

BUSCO runs & results:

Flye assembly
Command:
```
busco \
  -i ~/flot_project/assemblies/flye/assembly.fasta \
  -l bacteria_odb10 \
  -o busco_flye \
  -m genome \
  -f
```
Results:
```
C:98.4% [S:98.4%, D:0.0%], F:1.6%, M:0.0%, n:124
122 Complete BUSCOs
122 Complete & single-copy
0 Complete & duplicated
2 Fragmented
0 Missing
```

Raven assembly
Command:
```
busco \
  -i ~/flot_project/assemblies/raven/raven.fasta \
  -l bacteria_odb10 \
  -o busco_raven \
  -m genome \
  -f
```
Results:
```
C:99.2% [S:54.8%, D:44.4%], F:0.8%, M:0.0%, n:124
123 Complete BUSCOs
68 Complete & single-copy
55 Complete & duplicated
1 Fragmented
0 Missing
```

Shasta assembly
Command:
```
busco \
  -i ~/flot_project/assemblies/shasta/Assembly.fasta \
  -l bacteria_odb10 \
  -o busco_shasta \
  -m genome \
  -f
```
Results:
```
C:78.2% [S:78.2%, D:0.0%], F:5.6%, M:16.1%, n:124
97 Complete BUSCOs
97 Complete & single-copy
0 Complete & duplicated
7 Fragmented
20 Missing
```

Bandage Graphs (Flye)

```
conda create -y -n bandage_env -c bioconda bandage
conda activate bandage_env
```

Commands Used:

1. Generate PNG graphs from each assembler's GFA output:

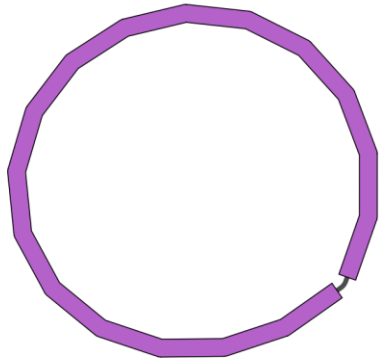
```bash
Bandage image ~/flot_project/assemblies/flye/assembly_graph.gfa docs/flye_graph.png
Bandage image ~/flot_project/assemblies/raven/raven.gfa docs/raven_graph.png
Bandage image ~/flot_project/assemblies/shasta/Assembly.gfa docs/shasta_graph.png
```

2. Verify PNG outputs and sizes:

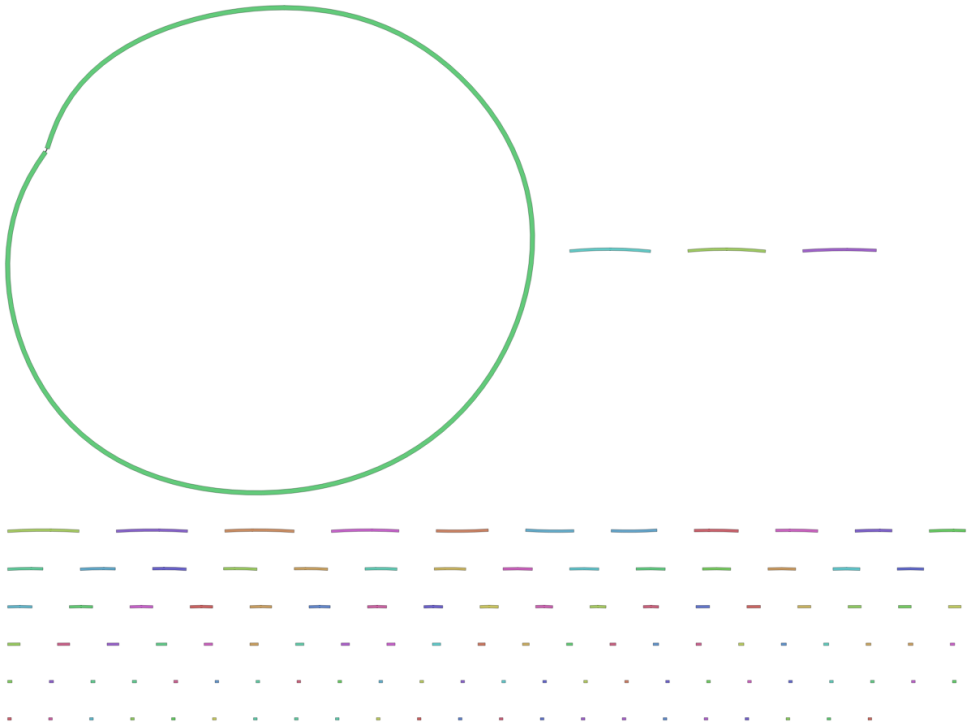
```bash
ls -lh docs/*.png
# Output:
-rw-r--r-- 1 abane_ashu abane_ashu  77K Aug 14 11:53 docs/flye_graph.png
-rw-r--r-- 1 abane_ashu abane_ashu  66K Aug 14 11:53 docs/raven_graph.png
-rw-r--r-- 1 abane_ashu abane_ashu 9.3K Aug 14 11:53 docs/shasta_graph.png
```

3. Open PNGs in Windows for inspection:

```bash
explorer.exe docs/flye_graph.png
explorer.exe docs/raven_graph.png
explorer.exe docs/shasta_graph.png
```



Raven vs Shasta



16S Extraction

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```
**1. Environment Setup**
We created a dedicated environment to avoid package conflicts and ensure reproducibility:
```bash
conda create -y -n ssu_env -c bioconda barrnap bedtools seqkit blast mafft trimal iqtree entrez-direct
conda activate ssu_env
```
- **barrnap**: Predicts rRNA genes (5S, 16S, 23S) using HMM models.
- **bedtools**: Extracts sequences from assemblies using genomic coordinates.
- **seqkit**: Sorts, filters, and manipulates FASTA/FASTQ files.
- **blast**: Compares sequences against NCBI databases.
- **mafft**: Performs multiple sequence alignment.
- **trimal**: Removes poorly aligned regions to improve phylogenetic accuracy.
- **iqtree**: Infers maximum likelihood phylogenetic trees.
- **entrez-direct**: Fetches sequences directly from NCBI.

2. 16S rRNA Detection with barrnap
We used barrnap to scan assemblies for bacterial 16S rRNA genes and output GFF files:
```bash
mkdir -p ~/flot_project/eval/16s
barrnap --kingdom bac ~/flot_project/assemblies/flye/assembly.fasta > ~/flot_project/eval/16s/flye.gff
barrnap --kingdom bac ~/flot_project/assemblies/raven/raven.fasta > ~/flot_project/eval/16s/raven.gff
barrnap --kingdom bac ~/flot_project/assemblies/shasta/Assembly.fasta > ~/flot_project/eval/16s/shasta.gff
```
We then filtered the GFF files to keep only 16S rRNA hits:
```bash
awk '$3=="rRNA" && /product=16S/' ~/flot_project/eval/16s/flye.gff > ~/flot_project/eval/16s/flye_16S.gff
awk '$3=="rRNA" && /product=16S/' ~/flot_project/eval/16s/raven.gff > ~/flot_project/eval/16s/raven_16S.gff
awk '$3=="rRNA" && /product=16S/' ~/flot_project/eval/16s/shasta.gff > ~/flot_project/eval/16s/shasta_16S.gff
```

3. Extracting 16S Sequences
We used bedtools to extract nucleotide sequences for the predicted 16S rRNAs:
```bash
bedtools getfasta -fi ~/flot_project/assemblies/flye/assembly.fasta -bed ~/flot_project/eval/16s/flye_16S.gff -s > ~/flot_project/eval/16s/flye_16S.fasta
bedtools getfasta -fi ~/flot_project/assemblies/raven/raven.fasta -bed ~/flot_project/eval/16s/raven_16S.gff -s > ~/flot_project/eval/16s/raven_16S.fasta
bedtools getfasta -fi ~/flot_project/assemblies/shasta/Assembly.fasta -bed ~/flot_project/eval/16s/shasta_16S.gff -s > ~/flot_project/eval/16s/shasta_16S.fasta
```
We sorted sequences by length and kept only the longest per assembler:
```bash
for ASM in flye raven shasta
do
  seqkit sort -l -r ~/flot_project/eval/16s/${ASM}_16S.fasta | seqkit head -n 1 > ~/flot_project/eval/16s/${ASM}_16S_rep.fasta
done
```
```

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```
**4. BLASTN Species Identification**
We ran BLASTN for each representative 16S sequence against the NCBI nt database:
```bash
for ASM in flye raven shasta
do
  blastn -query ~/flot_project/eval/16s/${ASM}_16S_rep.fasta -db nt -remote \
  -entrez_query "Bacteria[ORGN] AND 16S[Title]" \
  -max_target_seqs 10 -evalue 1e-50 \
  -outfmt "6 qseqid sacc pident length qlen qcovs bitscore stitle" \
  > ~/flot_project/eval/16s/${ASM}_blastn.tsv
done
```
Explanation of key parameters:
- **-entrez_query**: Restricts search to bacterial 16S sequences.
- **-max_target_seqs 10**: Limits results to top 10 hits.
- **-evalue 1e-50**: Ensures matches are highly significant.
- **-outfmt 6**: Outputs tabular format with specific columns.
Inspect Top Hit for Each Assembly
Command:
```bash
for ASM in flye raven shasta; do
  head -n 1 ~/flot_project/eval/16s/${ASM}_blastn.tsv
done
```
Output:
```text
Flye:   Acinetobacter baumannii (100.000% identity, 1529/1533 bp, strain 16842)
Raven:  Klebsiella pneumoniae (99.610% identity, 1537/1537 bp, strain NUBRI-K)
Shasta: Acinetobacter baumannii (99.804% identity, 1530/1534 bp, strain 16842)
```
** Building the Phylogenetic Tree**
We retrieved reference sequences for the top 15 Flye BLASTN hits:
```bash
cut -f2 ~/flot_project/eval/16s/flye_blastn.tsv | head -n 15 | paste -sd, - | \
xargs -I{} sh -c 'efetch -db nucleotide -id {} -format fasta' > ~/flot_project/eval/16s/ref_seqs.fasta
```
We combined assembler sequences with reference sequences:
```bash
cat ~/flot_project/eval/16s/*_16S_rep.fasta ~/flot_project/eval/16s/ref_seqs.fasta > ~/flot_project/eval/16s/all_16S.fasta
```
We aligned sequences with MAFFT and trimmed with trimAl:
```bash
mafft --auto --thread $(nproc) ~/flot_project/eval/16s/all_16S.fasta > ~/flot_project/eval/16s/all_16S.aln.fasta
trimal -automated1 -in ~/flot_project/eval/16s/all_16S.aln.fasta -out ~/flot_project/eval/16s/all_16S.trim.fasta
```
```

Phylogenetic Analysis

abane_ashu@AbaneAshu: ~/flot_project/docs

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```
We built the tree with IQ-TREE:
```bash
iqtree -s ~/flot_project/eval/16s/all_16S.trim.fasta -m MFP -bb 1000 -alrt 1000 -nt AUTO
```
We used the .contree file (consensus tree with bootstrap support) for visualization in iTOL:
```bash
cp ~/flot_project/eval/16s/all_16S.trim.fasta.contree ~/flot_project/eval/16s/16S_tree.nwk
```

We then downloaded the 16S_tree.nwk file and uploaded it to ITOL (Interactive Tree of Life), to visualise and manipulate the phylogenetic tree and create a PDF file of the phylogenetic relationship between all
```bash
cp ~/flot_project/eval/16s/16S_tree.nwk /mnt/c/Users/Louis/Downloads/
```

Checking Top BLASTN Matches (First 15 Hits)

Before building the phylogenetic tree, we wanted to examine the **first 15 BLASTN hits** for each assembler's representative 16S rRNA sequence.
This step helps us:
- Identify the most likely species matches.
- See whether the same or different bacterial species appear across assemblies.
- Select relevant reference sequences for phylogenetic tree construction.

Command used:
```bash
head -n 15 ~/flot_project/eval/16s/*_blastn.tsv
```
Explanation:
- **head -n 15**: Shows the first 15 lines of each BLASTN result file.
- This allows quick inspection of species names, accession numbers, and % identity.

Output:
```text
==> flye_blastn.tsv <==
contig_1:1206335-1207868(+)	HE651910	100.000 1529	1533	100	2824	Acinetobacter baumannii 16S rRNA gene, strain 16842
contig_1:1206335-1207868(+)	6YPU_2	99.870	1533	1533	100	2820	Chain 2, 16S ribosomal RNA
contig_1:1206335-1207868(+)	MT793124	99.870	1533	1533	100	2820	Acinetobacter baumannii strain AB1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	NR_117620	99.869	1529	1533	100	2813	Acinetobacter baumannii strain ATCC 19606 16S ribosomal RNA, partial sequence
contig_1:1206335-1207868(+)	EU883589	99.869	1529	1533	100	2813	Acinetobacter baumannii strain AIIMS 11 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	KT964444	99.804	1529	1533	100	2808	Acinetobacter baumannii strain HNHB02 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	KM408612	99.804	1529	1533	100	2808	Acinetobacter baumannii strain LJ17 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	MH393496	99.804	1528	1533	100	2806	Acinetobacter baumannii strain FOP220 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	PQ722327	99.869	1525	1533	99	2806	Acinetobacter baumannii strain Aci_bau1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)	PQ721429	99.869	1525	1533	99	2806	Acinetobacter baumannii strain Ac5 16S ribosomal RNA gene, partial sequence

==> raven_blastn.tsv <==
```

Activate Windows
Go to Settings to activate Windows.
```

Phylogenetic Tree Analysis and Comparison

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```
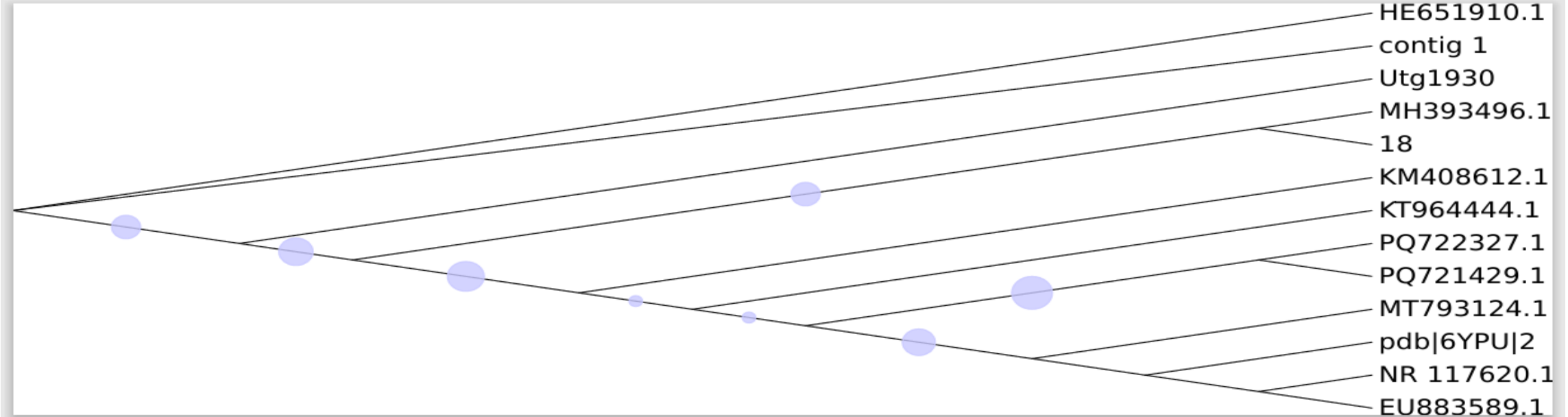
This allows quick inspection of species names, accession numbers, and % identity.

Output:
```text
==> flye_blastn.tsv <==
contig_1:1206335-1207868(+)     HE651910        100.000 1529    1533    100     2824    Acinetobacter baumannii 16S rRNA gene, strain 16842
contig_1:1206335-1207868(+)     6YPU_2  99.870  1533    1533    100     2820    Chain 2, 16S ribosomal RNA
contig_1:1206335-1207868(+)     MT793124        99.870  1533    1533    100     2820    Acinetobacter baumannii strain AB1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     NR_117620       99.869  1529    1533    100     2813    Acinetobacter baumannii strain ATCC 19606 16S ribosomal RNA, partial sequence
contig_1:1206335-1207868(+)     EU883589        99.869  1529    1533    100     2813    Acinetobacter baumannii strain AIIMS 11 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     KT964444        99.804  1529    1533    100     2808    Acinetobacter baumannii strain HNHB02 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     KM408612        99.804  1529    1533    100     2808    Acinetobacter baumannii strain LJ17 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     MH393496        99.804  1528    1533    100     2806    Acinetobacter baumannii strain FOP220 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     PQ722327        99.869  1525    1533    99      2806    Acinetobacter baumannii strain Aci_bau1 16S ribosomal RNA gene, partial sequence
contig_1:1206335-1207868(+)     PQ721429        99.869  1525    1533    99      2806    Acinetobacter baumannii strain Ac5 16S ribosomal RNA gene, partial sequence

==> raven_blastn.tsv <==
Utg1930:37498-39035(-)  MN429133        99.610  1537    1537    100     2804    Klebsiella pneumoniae strain NUBRI-K 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  PQ865839        99.610  1537    1537    100     2804    Uncultured Klebsiella sp. clone K-pneumoniae-1 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  PP340960        99.610  1537    1537    100     2804    Klebsiella sp. strain SG01 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  AB642255        99.674  1533    1537    100     2802    Klebsiella pneumoniae gene for 16S rRNA, partial sequence
Utg1930:37498-39035(-)  MZ401370        99.545  1537    1537    100     2798    Klebsiella pneumoniae strain M27 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401365        99.545  1537    1537    100     2798    Klebsiella pneumoniae strain M3 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  JN644536        99.738  1526    1537    99      2795    Klebsiella pneumoniae strain BR10_2A 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401368        99.480  1537    1537    100     2793    Klebsiella pneumoniae strain M20 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  MZ401364        99.480  1537    1537    100     2793    Klebsiella pneumoniae strain M2 16S ribosomal RNA gene, partial sequence
Utg1930:37498-39035(-)  AB642256        99.543  1533    1537    100     2791    Klebsiella pneumoniae gene for 16S rRNA, partial sequence

==> shasta_blastn.tsv <==
18:21581-23115(+)       HE651910        99.804  1530    1534    100     2808    Acinetobacter baumannii 16S rRNA gene, strain 16842
18:21581-23115(+)       6YPU_2  99.674  1534    1534    100     2804    Chain 2, 16S ribosomal RNA
18:21581-23115(+)       MT793124        99.674  1534    1534    100     2804    Acinetobacter baumannii strain AB1 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       KM408612        99.739  1530    1534    100     2802    Acinetobacter baumannii strain LJ17 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       MH393496        99.738  1529    1534    100     2800    Acinetobacter baumannii strain FOP220 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       EU078177        99.738  1529    1534    100     2800    Acinetobacter baumannii strain BD5 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       NR_117620       99.673  1530    1534    100     2796    Acinetobacter baumannii strain ATCC 19606 16S ribosomal RNA, partial sequence
18:21581-23115(+)       EU883589        99.673  1530    1534    100     2796    Acinetobacter baumannii strain AIIMS 11 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       KT964444        99.608  1530    1534    100     2791    Acinetobacter baumannii strain HNHB02 16S ribosomal RNA gene, partial sequence
18:21581-23115(+)       PQ722327        99.672  1526    1534    99      2789    Acinetobacter baumannii strain Aci_bau1 16S ribosomal RNA gene, partial sequence
```


```

Prokka(Annotation) + BlastKoala For Metabolic Analysis

abane_ashu@AbaneAshu: ~/flot_project/docs

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```
### 1. Environment Setup
We created a dedicated Conda environment to ensure Prokka ran with all necessary dependencies without conflicting with other tools in our workflow:
conda create -n annotation_env -c bioconda -c conda-forge prokka
conda activate annotation_env
This isolated environment guaranteed reproducibility and minimized dependency issues.

### 2. Running Prokka
Prokka is a rapid prokaryotic genome annotation tool. It predicts coding sequences (CDS), rRNAs, tRNAs, and assigns functional annotations by comparing against curated databases.
We annotated the assemblies from all three assemblers:

# Flye
prokka ~/flot_project/assemblies/flye/assembly.fasta \
  --outdir ~/flot_project/eval/annotation/flye_prokka \
  --prefix flye

# Raven
prokka ~/flot_project/assemblies/raven/assembly.fasta \
  --outdir ~/flot_project/eval/annotation/raven_prokka \
  --prefix raven

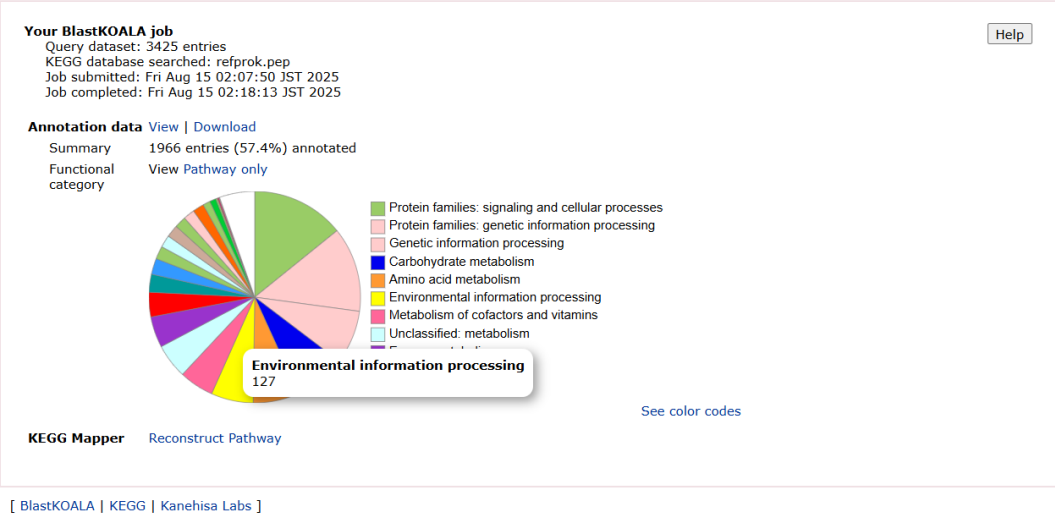
# Shasta
prokka ~/flot_project/assemblies/shasta/assembly.fasta \
  --outdir ~/flot_project/eval/annotation/shasta_prokka \
  --prefix shasta

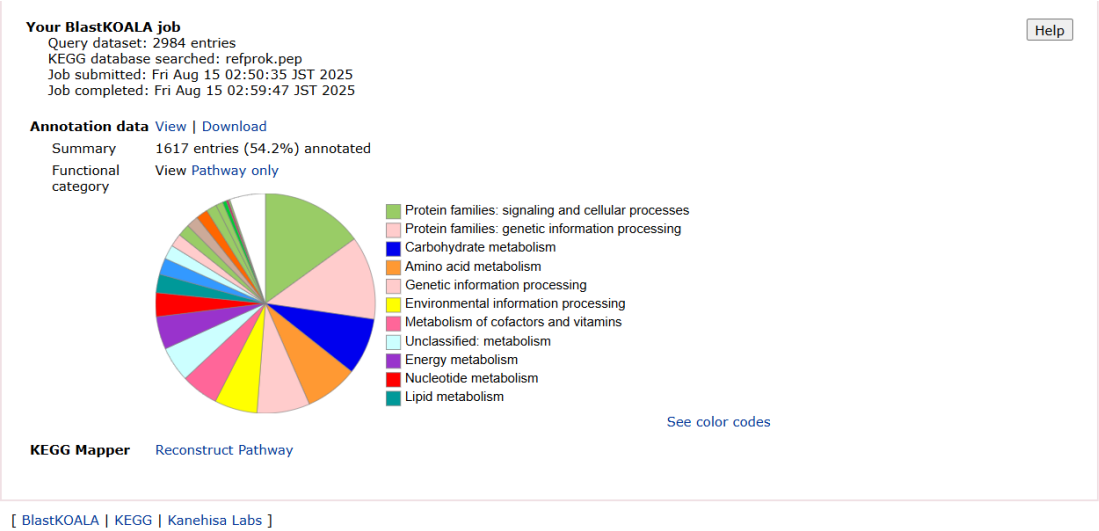
### 3. Key Output: The .faa Files
From Prokka, the most important output for this step was the .faa file:
- This file contains all predicted protein sequences in FASTA format.
- Proteins are the functional workhorses of the cell, and their sequences allow us to link genome data to metabolic pathways.
- KEGG's BlastKOALA uses these sequences to assign KEGG Orthology (KO) numbers and map them to metabolic pathways.

### 4. Preparing Files for BlastKOALA
We copied the .faa files from our Linux environment to our local machine to upload them to BlastKOALA:
cp ~/flot_project/eval/annotation/flye_prokka/flye.faa /mnt/c/Users/Louis/Downloads/
cp ~/flot_project/eval/annotation/raven_prokka/raven.faa /mnt/c/Users/Louis/Downloads/
cp ~/flot_project/eval/annotation/shasta_prokka/shasta.faa /mnt/c/Users/Louis/Downloads/
The /mnt/c/... path bridges the WSL/Linux environment to the Windows filesystem, placing the files into our Downloads folder for easy web upload.

### 5. Running BlastKOALA
We uploaded each .faa file to KEGG's BlastKOALA (http://www.kegg.jp/blastkoala/) using the 'Prokaryotes' database (refprok.pep).
BlastKOALA matches protein sequences to KO terms and groups them into functional categories such as metabolism, genetic information processing, and environmental processing.
```

BlastKoala Results





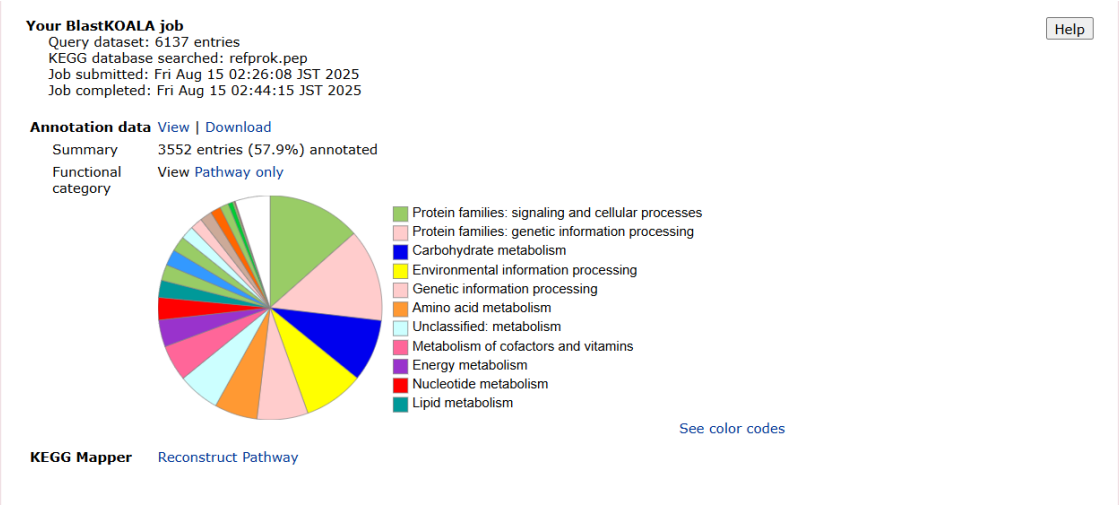


Fig A: Acinobacter baumanii (FLYE)- proteins predicted is 57.5%

Fig B: Klebsiella pneumoniae (RAVEN)- Proteins predicted is 57.9%

Fig C: Acinobacter baumanii (SHASTA)- Proteins predicted is 54.2%

CONCLUSION

- This BlastKOALA metabolic analysis forms the final, complementary layer of our project, linking structural genome data to biologicaion.
- From assembly to metabolic annotation, our pipeline demonstrated:

- Flye **consistently** provided the most **complete and contiguous genome**, yielding the richest functional annotation for A. baumannii.
- Shasta, although fragmented and incomplete, still converged with Flye at both the 16S phylogenetic and metabolic profile levels, proving the resilience of conserved gene regions.
- Raven, despite excellent completeness scores from KAT and BUSCO, diverged in both phylogeny and metabolic signature, underlining that completeness alone does not ensure biological accuracy.

CONCLUSION

- Our analysis integrated assembly metrics QUAST, KAT, BUSCO, Bandage, taxonomic inference (BLASTN + phylogeny), and functional capacity (Prokka + BlastKOALA),
- We achieved a holistic view of each assembler's performance technically and biologically.
- This final step confirms that genome assembly evaluation should always extend beyond structural metrics to include functional and metabolic perspectives for robust, real-world conclusions.